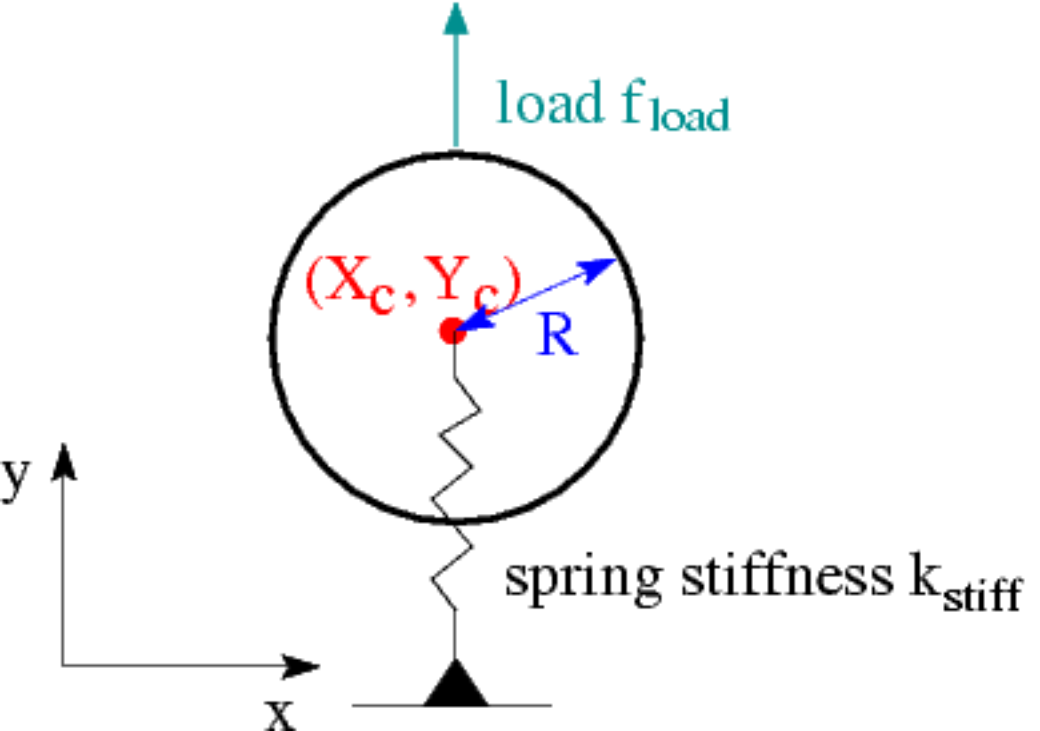
load $f_{load}$
$(X_c, Y_c)$
R
y
x
spring stiffness $k_{stiff}$